WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – II : HYDROGRAPH ANALYSIS & GROUND WATER

I. Hydrograph

> A hydrograph is a graph of discharge (Q) versus time (t) at a given section of a stream. It integrates all catchment characteristics and the storm that caused the flow.

- Types: annual hydrograph (daily/ monthly flows – perennial or intermittent streams), flood (storm) hydrograph (single storm), hourly hydrograph; unit hydrograph; synthetic hydrograph.

❖Components of a flood hydrograph

- Rising limb (concentration curve): discharge increasing; shape depends on storm intensity & catchment, antecedent conditions.
- Crest segment (peak): maximum discharge Q_p, occurs after time to peak t_p.
- Falling/ recession limb: decreasing flow after rain ends; controlled by catchment storage (depletion curve).
- Base flow: ground-water contribution. Direct runoff = surface runoff + interflow.
- Basin lag (t_l): time from centroid of rainfall to peak of hydrograph. Time base T_b: total duration of direct runoff. Time of concentration t_c.

❖Factors affecting shape

- Catchment area, shape (fan vs elongated), slope, drainage density, soil/ land use, storm intensity, duration, movement; urbanisation → higher & earlier peak.

❖Recession equation

$$Q_t = Q_0 K_r^t \text{ or } Q_t = Q_0 e^{-at} \quad (K_r = \text{recession constant} < 1)$$

2. Unit Hydrograph (UH) – Sherman 1932

> The unit hydrograph of a catchment is the direct-runoff hydrograph resulting from 1 cm (unit depth) of net (effective) rainfall, uniformly distributed over the basin, occurring at a uniform rate for a specified duration D hours. Called D-hour unit hydrograph.

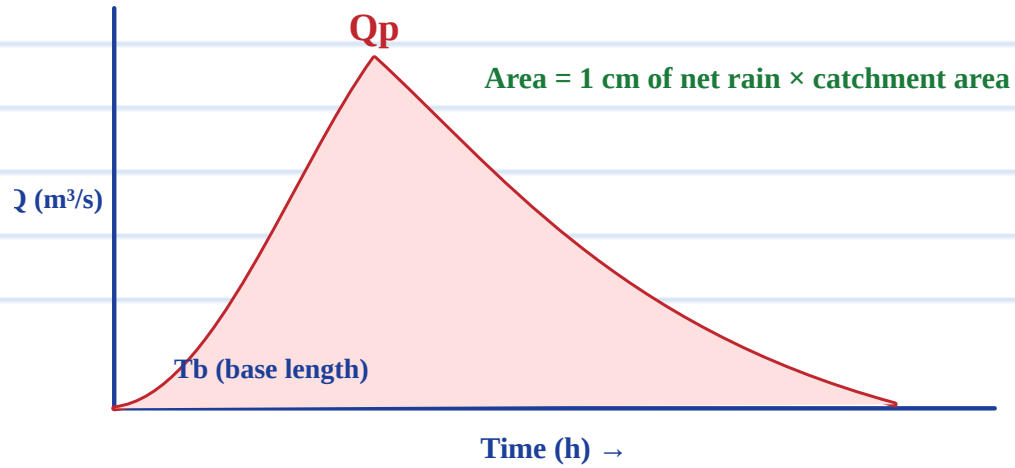


Fig 2.1 Unit hydrograph (UH) of duration D hours

❖Basic assumptions

- 1. Constant (uniform) intensity of effective rainfall for duration D.
- 2. Uniform areal distribution over the entire catchment.
- 3. Time invariance: same hydrograph for same effective rain duration irrespective of when it occurs.
- 4. Linearity (superposition): ordinates of DRH are directly proportional to effective rainfall; hydrographs of successive rains can be added (principle of proportionality & superposition).
- 5. Time base of direct runoff is constant for rains of equal duration.

❖Properties

- Area under UH = volume = 1 cm × A (catchment area).
- Ordinates scale with rain depth: DRH ordinates for r cm = r × UH ordinates.
- Peak flow and time to peak depend on D.

3. Construction (Derivation) of Unit Hydrograph from a Storm Hydrograph

❖Steps

- 1. Select an isolated, simple, uniform-intensity storm of duration D with 1–5 cm effective rain.
- 2. Obtain stream-flow hydrograph; separate base flow (straight line/ fixed base).
- 3. Calculate direct runoff volume; depth of direct runoff = volume/ catchment area (= net rainfall R_e in cm).
- 4. Divide DRH ordinates by R_e → ordinates of D-h UH.
- 5. Check area under UH = 1 cm (volume = A × 0.01 m).
- 6. Plot UH vs time.

❖Example – derivation of 6-h UH

> Catchment A = 36 km^2; total hydrograph & base flow (m^3/s):

t (h)	0	6	12	18	24	30	36
Total flow	10	50	120	90	50	25	10
Base flow	10	10	10	10	10	10	10
DRH	0	40	110	80	40	15	0

$\Sigma DRH = 285\ m^3/s$ → volume = $285 \times 6 \times 3600 = 6.156 \times 10^6\ m^3$; $R_e = 6.156\times10^6/(36\times10^6) = 0.171\ m = 17.1$ cm. UH ordinates = DRH/17.1 → 0, 2.34, 6.43, 4.68, 2.34, 0.88, 0 m^3/s (check area ≈ 1 cm).

4. Application – Flood Hydrograph from Unit-Duration Rainfall

❖Using UH to get DRH for a given storm (convolution)

- Given: D-hour UH ordinates $U_1, U_2, \ldots$; storm with effective rainfall depths $R_1, R_2, \ldots$ in successive D-hour intervals.
- Principle: DRH from each block = R_j × UH, shifted by D hours for each next block; add (superpose) ordinates.

$$Q_n = \Sigma_{m=1}^{n} R_m \cdot U_{n-m+1} \text{ (discrete convolution)}$$

- Then add base flow to get total flood hydrograph.

❖Example

> 2-block storm: effective rain 2 cm (first 6 h), 3 cm (next 6 h). 6-h UH ordinates (at 6-h steps): 0, 5, 20, 15, 8, 3, 0.

- 2 cm hydrograph: 0, 10, 40, 30, 16, 6, 0, 0
- 3 cm hydrograph (lagged 1 step): 0, 0, 15, 60, 45, 24, 9, 0
- DRH = sum: t=0: 0; 6: 10; 12: 55; 18: 90; 24: 61; 30: 30; 36: 9; 42: 0. Peak = 90 m^3/s at 18 h. Add base flow (say 10) → total peak 100 m^3/s.

❖Other uses of UH

- Flood forecasting & design flood (from design storm), reservoir routing inputs, effect of land-use change, synthetic UH for ungauged basins (Snyder's method: $t_p = C_t (L L_{ca})^{0.3}$; SCS triangular UH: $Q_p = 2.08\ A / t_p$, with A in km^2, Q in m^3/s per cm).

❖Limitations of the unit hydrograph

- Applicable only to catchments of 25 – 5000 km^2 (very small: rational; very large: non-uniform rain, channel routing needed).
- Assumes uniform rainfall distribution & intensity, which rarely occurs; non-linear behaviour in large floods & very small ones.
- Requires accurate gauge data; difficult to separate base flow; snow-melt effects & storage reservoirs not handled.
- Catchment change (urbanisation) invalidates; rainfall duration must be near D (D ≈ 1/4 of basin lag).
- Not applicable to areas with large temporary storage or ungauged (needs synthetic UH).

5. S-Hydrograph (S-Curve)

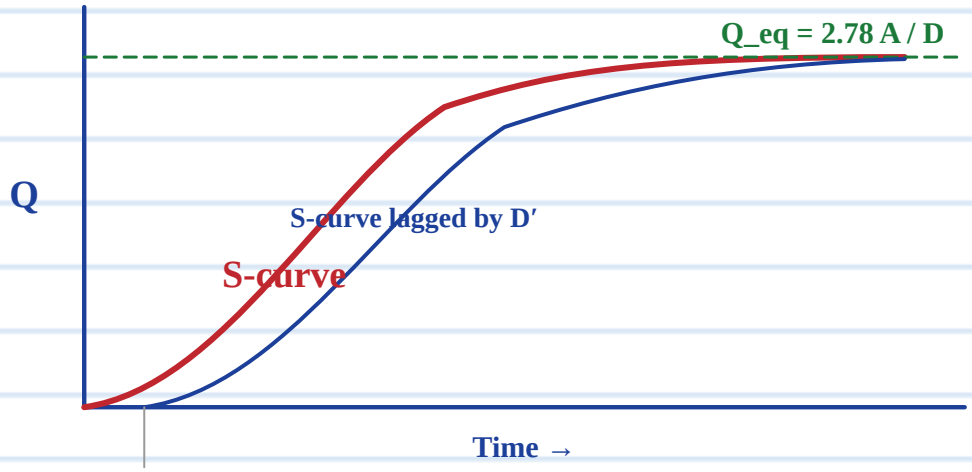


Fig 2.2 S-hydrograph: summation of continuous series of D-h unit hydrographs

S-hydrograph = hydrograph produced by a **continuous effective rainfall at a constant rate of 1 cm/ D hours** (i.e., intensity 1/D cm/h) over an **indefinite period**; obtained by **summing D-hour UHs, each lagged by D hours.**

- Shape: rises like an "S" and **becomes constant at equilibrium discharge Q_e** when inflow = outflow:

$$Q_e = 2.78\ A / D\ (m^3/s;\ A \text{ in } km^2,\ D \text{ in hours})\ [\text{since } Q_e = A \times (1\ cm / D)]$$

❖Construction

- 1. Write the D-h UH ordinates at intervals D. 2. Lag each ordinate series by D, 2D, 3D, ... 3. **Sum** of all → S-curve ordinates S(t) (equilibrium after about T_b/D UHs). 4. Smooth; check Q_e = 2.78 A/D.

❖Use – derive a UH of different duration D'

- Lag the S-curve by D' hours, **subtract** from original: S(t) – S(t – D') = hydrograph for intensity (1/D) acting D' hours = (D'/D) cm of rain.

$$D'\text{-h UH ordinates} = (D / D')\ [S(t) - S(t - D')]$$

- Eg. from 6-h UH get 3-h UH or 12-h UH. Also used when D' is not a multiple of D (not possible by direct lagging).
- If S-curve oscillates (non-uniform data) → hump; smooth it.

❖Example

6-h UH ordinates: 0, 5, 20, 15, 8, 3, 0. S-curve: 0, 5, 25, 40, 48, 51, 51 (equilibrium 51 → $A = 51 \times 6/2.78 = 110\ km^2$). 12-h UH (D' = 12, D = 6): ordinates (S(t) − S(t−12))×(6/12): t=6: 5/2 = 2.5; t=12: (25−0)/2 = 12.5; t=18: (40−5)/2 = 17.5; t=24: (48−25)/2 = 11.5; t=30: (51−40)/2 = 5.5; t=36: (51−48)/2 = 1.5.

6. GROUND WATER – Introduction

Ground water = water **below the ground surface in the saturated zone** (pores/ fissures of soil & rock); ~ 30% of fresh water, ~ 60% of India's irrigation & 85% of rural drinking water. Recharged by **rain infiltration, rivers, canals, tanks, irrigation return flow, artificial recharge.**

- **Advantages:** reliable (less evaporation), local, good quality, drought buffer. **Problems:** over-exploitation (falling water table in A.P., Punjab, Gujarat), fluoride/ arsenic/ salinity, seawater intrusion.
- **Study includes:** occurrence, movement, exploration, well hydraulics, recharge, quality.

7. Aquifer, Aquiclude, Aquifuge & Aquitard



Fig 2.3 Sub-surface water zones and aquifer types

- **Aquifer:** geologic formation that stores and transmits water in sufficient quantity to wells/ springs – sand, gravel, weathered or fractured rock, limestone, sandstone. (Good porosity and permeability.)

- **Aquiclude:** formation that stores water (porous) but does not transmit it appreciably – clay, shale (high porosity, very low permeability).
- **Aquifuge:** formation neither stores nor transmits water – solid granite/ unfractured rock/ compact basalt.
- **Aquitard:** poorly permeable formation (silty clay) that transmits water slowly – allows leakage between aquifers (leaky layer).

❖Summary

Term	Stores water?	Transmits water?	Example
Aquifer	Yes	Yes	Sand, gravel
Aquiclude	Yes	No	Clay
Aquifuge	No	No	Massive granite
Aquitard	Yes	Very slowly	Sandy clay

8. Aquifer Parameters

❖(a) Porosity (n)

Porosity = ratio of volume of voids (pores) to total volume of the soil/ rock mass.

$$n = V_v / V \times 100\ \% \text{ ; relation with void ratio e: } n = e / (1 + e)$$

- Typical: clay 40–55%, silt 35–50%, sand 25–40%, gravel 25–40%, sandstone 5–30%, fractured granite 1–10%.
- **Primary** (original pores) & **secondary** (fractures, solution cavities) porosity. **Effective porosity** = interconnected pores (what yields water).

❖(b) Specific yield (S_y) and (c) Specific retention (S_r)

Specific yield = ratio of the volume of water that drains by gravity from a saturated soil to the total volume of soil. **Specific retention** = ratio of volume of water retained against gravity (by molecular & capillary forces) to total volume.

$$n = S_y + S_r \text{ ; } S_y = V_{drained}/V \text{ ; } S_r = V_{retained}/V$$

Material	n %	S_y %	S_r %
Clay	45	3	42
Silt	45	8	37
Fine sand	35	21	14
Coarse sand	30	27	3
Gravel	25	22	3

- S_y decides the usable storage in unconfined aquifers; coarse materials have high S_y, clay high S_r.
- Volume of water released from an aquifer of area A with water-table fall Δh: $V = S_y A \Delta h$.

❖Example

> Sample of 1000 cm^3 saturated sand yields 250 cm^3 by drainage and retains 100 cm^3: S_y = 25%, S_r = 10%, n = 35%.

> Water table falls 2 m over 5 km^2, S_y = 0.15: $V = 0.15 \times 5 \times 10^6 \times 2 = 1.5 \times 10^6$ m^3.

9. Divisions of Sub-surface Water

> Sub-surface water is divided vertically into two zones by the **water table**:

❖1. Zone of aeration (unsaturated/ vadose zone) – pores partly water, partly air

- (i) **Soil-water (root) zone**: near ground, used by plants; capillary & hygroscopic water.
- (ii) **Intermediate (vadose) zone**: between root zone and capillary fringe; water held by hygroscopic & capillary forces; gravity water passes through.
- (iii) **Capillary fringe (zone)**: just above the water table; water raised by **capillary action** (height $h_c = 0.15/d$ cm; clay ≈ 2–4 m, sand ≈ 0.1–0.5 m).

❖2. Zone of saturation (phreatic zone)

- All pores filled with water; water here is **ground water**; its upper surface = **water table** (pressure = atmospheric).

❖Types of subsurface water

- Hygroscopic, capillary, gravitational (free) water, combined/ chemically bound, juvenile (magmatic), connate (trapped in sediments), meteoric.

10. Water Table

> Water table (phreatic surface) = the upper surface of the zone of saturation, where the water pressure = atmospheric; level observed in an open well.

- Shape: follows the topography (subdued); slopes toward streams/ wells; cone of depression around a pumping well; rises after rains/ recharge, falls in summer & by pumping; fluctuations measured by observation wells/ piezometers.
- Piezometric (potentiometric) surface – imaginary surface to which water rises in a well in a confined aquifer; may be above the ground → artesian (flowing) well.
- Depth of water table: deep (> 8 m) → favourable; shallow (< 2 m) → water-logging.
- Perched water table: local saturated layer above an aquiclude lens, above the main water table.
- Uses: direction of ground-water flow, recharge/ discharge estimate, depth to drill wells.

11. Types of Aquifers

- 1. Unconfined (water-table/ phreatic) aquifer: upper boundary = water table, open to atmosphere; recharged directly from surface; S_y = 0.01 – 0.30; well water stands at water table.
- 2. Confined (artesian/ pressure) aquifer: sandwiched between two impermeable layers (aquicludes); water under pressure > atmospheric; recharge area at outcrop far away; storage coefficient 10^{-5} – 10^{-3}; wells may be flowing artesian.
- 3. Semi-confined (leaky) aquifer: one or both confining layers are aquitards allowing leakage.
- 4. Perched aquifer: small unconfined body held above main water table by a lens of impermeable material.
- 5. Others: alluvial (river sands – high yield), hard-rock (fractured/ weathered granite, basalt) – typical of A.P. & Telangana; coastal (salinity), karstic limestone.

❖Unconfined vs confined

Feature	Unconfined	Confined
Upper boundary	Water table	Impermeable layer
Pressure	Atmospheric	> Atmospheric
Storage coefficient	0.01–0.30 ($\approx S_y$)	10^{-5}–10^{-3}
Water released by	Dewatering pores	Compression of aquifer & expansion of water
Recharge	Local rainfall	Distant outcrop
Well	Normal	Artesian (may flow)

12. Storage Coefficient (Storativity, S)

> Storage coefficient = volume of water an aquifer releases from (or takes into) storage per unit surface area of aquifer per unit change in head (h) normal to that surface (dimensionless).

$$S = \Delta V / (A \cdot \Delta h)$$

- Unconfined: $S \approx S_y$ (0.01–0.30) as water comes from drainage of pores.
- Confined: $S = \rho\, g\, b\, (\alpha + n\beta) \approx 10^{-5} - 10^{-3}$ (b = thickness, α = aquifer compressibility, β = water compressibility) – tiny because water comes from elastic compression.
- Specific storage $S_s = S/b$ (per metre).
- Determined by pumping tests (Theis method) with observation wells.

13. Coefficient of Permeability (Hydraulic Conductivity, K)

> Darcy's law: velocity of flow through a saturated porous medium is proportional to hydraulic gradient: $v = K i$, $i = dh/dL$.

$$Q = K i A = K (\Delta h / L) A \ (m^3/s) ; K = [\text{m/s or m/day}]$$

- Coefficient of permeability K = discharge per unit area per unit hydraulic gradient (velocity under unit gradient) – depends on soil (grain size) and fluid (viscosity, density): $K = k (\rho\, g / \mu)$, k = intrinsic permeability.
- Range: gravel 10^{-1}–10^{1} cm/s; clean sand 10^{-3}–10^{-1}; silty sand 10^{-5}–10^{-3}; silt 10^{-6}–10^{-4}; clay < 10^{-7} cm/s.
- Hazen's formula: $K = C\, d_{10}^{\,2}$ (cm/s, d_{10} in mm; $C \approx 100$–150).
- Seepage velocity $v_s = v/n$ (n = porosity).

- Determination: constant-head test (coarse), falling-head test (fine soil): $K = (aL/At)\ln(h_1/h_2)$, pumping test (field), tracer test.
- Validity of Darcy's law: laminar flow, Reynolds number $Re < 1 - 10$.
- Layered aquifer: equivalent horizontal $K_h = \Sigma K_i b_i / \Sigma b_i$; vertical $K_v = \Sigma b_i / \Sigma (b_i/K_i)$.

14. Transmissibility (Transmissivity, T)

> Transmissibility = rate of flow of water through a vertical strip of aquifer of unit width, extending full saturated thickness, under unit hydraulic gradient.

$$T = K\,b\ (m^2/day)\ ;\ Q = T\,i\,B\ (B = width)$$

- b = saturated thickness (unconfined: $b = h$, varies).
- T high ($> 1000\ m^2/day$) → excellent aquifers (alluvium); 100–1000 good; < 10 poor.

❖Well hydraulics (steady radial flow – Dupuit/ Thiem)

- Unconfined aquifer well: $Q = \pi K (h_2^2 - h_1^2) / \ln(r_2/r_1)$
- Confined aquifer well: $Q = 2\pi K b (h_2 - h_1) / \ln(r_2/r_1) = 2\pi T (h_2 - h_1) / \ln(r_2/r_1)$
- Also in terms of drawdowns s_1, s_2: $T = Q \ln(r_2/r_1) / [2\pi(s_1 - s_2)]$.
- Specific capacity = Q/s (m^3/h per m drawdown); radius of influence R (Sichardt: $R = 3000\, s\sqrt{K}$).
- Theis non-equilibrium: $s = (Q/4\pi T)\,W(u)$, $u = r^2S/(4Tt)$.

❖Solved problems

> P1. Sand aquifer $K = 20$ m/day, thickness $b = 15$ m: $T = 20 \times 15 = 300\ m^2/day$.

> P2. Seepage: $K = 10^{-3}$ cm/s $= 10^{-5}$ m/s, gradient 0.01, area $100\ m^2$: $Q = K\,i\,A = 10^{-5} \times 0.01 \times 100 = 1 \times 10^{-5}\ m^3/s = 0.864\ m^3/day$.

> P3. Confined aquifer $b = 20$ m, $K = 30$ m/day; drawdown heads: $h_1 = 18$ m at $r_1 = 10$ m, $h_2 = 19.5$ m at $r_2 = 100$ m: $Q = 2\pi \times 30 \times 20 \times (1.5)/\ln 10 = 5655/2.303 = 2456\ m^3/day$.

P4. Unconfined well: $K = 40$ m/day, $h_1 = 20$ m ($r = 0.3$ m), $h_2 = 25$ m ($r = 300$ m): $Q = \pi \times 40 \times (625 - 400)/\ln(1000) = 28274/6.908 = \mathbf{4093\ m^3/day}$.

P5. Porosity 0.4, $K = 1$ m/day, $i = 0.005 \rightarrow v = 0.005$ m/day; seepage velocity $= 0.005/0.4 = \mathbf{0.0125}$ m/day.

15. Important Questions – Unit II

- Define unit hydrograph; state assumptions & limitations. Derive a UH from a storm hydrograph.
- Explain construction of S-curve and derivation of UH of different duration; use in flood hydrograph.
- Distinguish aquifer, aquiclude, aquifuge. Porosity, specific yield, specific retention (with relation).
- Divisions of sub-surface water; water table; types of aquifers.
- Define storage coefficient, permeability and transmissibility; Darcy's law; well formulas.

Key relations: $n = Sy + Sr$; $T = K\,b$; $Qe = 2.78\,A/D$; $UH(D') = (D/D')\,[S(t) - S(t-D')]$